MILITARY SPECIFICATION

MUNITION FOR PROJECTILE, 155MM, H.E. M692/M731
LOADING AND ASSEMBLING OF

This amendment forms a part of Military Specification MIL-M-48186A(AR), dated 12 August 1977, and is approved for use by the U. S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Drawings, 9298401 - Plug Housing: Delete and add:

"9298401 - Plug Housing (alternate)".

PAGE 4

3.7: Delete the second sentence and substitute the following new sentence in its place:

"The bulk density of the explosive composition A5 loaded into the upper and lower shell assemblies shall be not less than 0.83 nor more than 1.00 grams/cc as determined by the "Bechtel Method".

PAGE 4

3.9: Delete paragraph and add the following:

"Molding compound for casting or assembling plugs into housings. The molding compound used for casting or assembling the plug into the housing shall comply with MIL-M-48245. The resin/hardener weight ratio shall be 2.0/1.0 ± 5% (1.8/1 to 2.2/1) and the ultimate compressive strength shall be 25,000 p.s.i. minimum for layers other than one (1) and five (5). A minimum compressive force of 27,000 p.s.i. shall have unrestricted use.

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PAGE 9

4.4.1.c: Delete in its entirety.

PAGE 12

4.4.2.3: Reword Note 1 as follows:

1. "Certification shall include the actual percentage by weight of each constituent immediately prior to blending."

PAGE 14

4.4.2.5: Delete and substitute with new 4.4.2.5, attached hereto.

PAGE 16

4.4.2.7: Delete and substitute with new 4.4.2.7, attached hereto.

PAGE 17

* 4.4.2.8: Delete and substitute with new 4.4.2.8, attached hereto.

PAGE 18

4.4.2.9: Delete and substitute with new 4.4.2.9, attached hereto.

PAGE 21

4.4.2.12: Delete and substitute with new 4.4.2.12, attached hereto.

PAGE 24

4.4.2.15: Delete and substitute with new 4.4.2.15, attached hereto.

PAGE 25

4.4.2.16, Major 101: Delete and substitute with new 4.4.2.16, attached hereto.

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PAGE 27

4.4.2.18: Delete and substitute with new 4.4.2.18, attached hereto.

PAGE 33

4.4.2.24: Delete title and substitute new 4.4.2.24, attached hereto.

PAGE 34

4.4.2.25: Delete title and substitute new title to read: "Plug, Housing (alternate)".

PAGE 35

4.4.2.26: Delete and substitute new 4.4.2.26, (Sheets 1 and 2), page numbers 35 and 35A, attached hereto.

PAGE 35A

4.4.2.27: Add new Page 35B, attached hereto.

PAGE 35B

4.4.2.28: Add new Page 35C, attached hereto.

PAGE 41

4.5.9, lines 1-10: Delete "The assemblies shall be temperature conditioned ... be made for evidence of propellant leakage."

PAGE 43

4.5.12: Change title to read:

"Quality controls for each molding compound batch for casting or assembling plugs into housings."

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4.5.14.1: Delete in its entirety and substitute the following:

"4.5.14.1: Test arena. The arena or room shall be not less than twenty (20) feet square and fifteen (15) feet in height. It shall be constructed of material that will resist the blast of twenty-five (25) grams of high explosive. The walls and ceiling shall be free of areas where fragments, parts, etc., could lodge and be lost. At least one wall shall be constructed to permit observation of functioning. A camera shall be positioned with a viewing port set at a height above the test surface to photograph item functioning in front of a suitable grid for subsequent examination and evaluation. Alternate construction may be used with the approval of the contracting officer.

Provisions shall be made to determine the velocity of the kill mechanism by determining the distance it travels and measuring the time from wedge breakup to the audible report from kill mechanism functioning."

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The attached insertable replacement pages listed below are replacements for the stipulated pages:

Paragraph No.	Old Page No.	New Page No.
4.4.2.5	14	14
4.4.2.7	16	16
4.4.2.8	17	17
4.4.2.9	18	18
4.4.2.12	21	21
4.4.2.15	24	24
4.4.2.16	25	25
4.4.2.18	27	27
4.4.2.24	33	33
4.4.2.26	35	35 and 35A
4.4.2.27	35A	35B
4.4.2.28	35B	35C

"The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the reqiurements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-AB14)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Propellant Nut Assembly (prior to assembly of orifice plate)	1 OF 1	9298433
			NEXT HIGHER ASSEMBLY 9298433

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Weight of propellant			3.2	Note A
Minor 201	Evidence of poor workmanship		0.65%	3.12	Visual
NOTE A	The Propellant Nut Assembly machine will be qualified at the beginning of each production interval by processing a minimum of two (2) acceptable weight samples prior to use in production. The machine will be requalified each time it is removed from the production process for maintenance of adjustment. A minimum of two (2) samples shall be processed each hour of production or fraction thereof. Each hour's production or fraction thereof shall be accepted provided all samples processed during that period meet weight requirements. If a sample fails to meet weight requirements, all loaded propellant nut assemblies processed since the last acceptable check shall be rejected. The assembly machine shall be stopped for corrective action and requalified prior to release for production.				

NOTES:

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Quality conformance inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186A(AR)
AMENDMENT 8

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Explosive Composition A5, prior to loading	1 OF 1	9298357
			NEXT HIGHER ASSEMBLY 9298357

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bulk density		---	3.7	4.5.2.1
102	Moisture content		---	3.7	4.5.1
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Upper Shell Assembly				9298357
					NEXT HIGHER ASSEMBLY 9298417
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Presence of sleeve, outer, positioned over tube assembly		100%	3.2	Visual
Major					
101	Sleeve height		0.40%	3.2	Gage
102	Depth to charge		0.40%	3.2	Gage
103	Outside diameter, max.		0.40%	3.2	Gage
104	Diameter of detonator tube sleeve, min		0.40%	3.2	Gage
105	Inside diameter of detonator tube		0.40%	3.2	Gage
106	Charge chipped, cracked or pitted		0.40%	3.2	Visual
107	Density of charge			3.2	4.5.2
108	Spring grip fastener missing		0.40%	3.2	Visual
109	Consolidation pressure			3.7	Certification
110	Presence of exuded adhesive (360°) at A5 interface (presence of RTV (360°) on top of sleeve		0.40%	3.2	Visual
111	Outersleeve compressive shear strength		Note 1	3.2	Gage
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES: Two (2) samples shall be prepared at the start and per every two (2) hours during a production shift; or whenever a new batch or lot or primer or adhesive is introduced.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186A(AR)
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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Lower Shell Assembly	1 OF 1	9298358
			NEXT HIGHER ASSEMBLY: 9298417

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth to charge		0.40%	3.2	Gage
102	Outside diameter, max		0.40%	3.2	Gage
103	Diameter of hole in shell		0.40%	3.2	Gage
104	Diameter of hole in charge		0.40%	3.2	Gage
105	Charge chipped, cracked or pitted		0.40%	3.2	Visual
106	Density of charge			3.2	4.5.2
107	Consolidation pressure			3.7	Certification
108	Comp. as shoulder, 360° not present		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Seal, Upper Shell (prior to assembly to kill mechanism)				9298418
					NEXT HIGHER ASSEMBLY 9298417
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pad missing		0.40%	3.2	Visual
102	Hole in pad missing		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48186(AR)
AMENDMENT 8

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.15	Spacer, Overlay	1	1	9298415
				NEXT HIGHER ASSEMBLY: 9298412-1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		0.40%	3.2	Gage
102	Largest outside diameter		0.40%	3.2	Gage
103	True position of largest outside diameter with hole		0.40%	3.2	Gage
104	Thickness over circular and rectangular protrusions (12 places)		0.40%	3.2	Gage
105	Thickness adjacent to protrusions		0.40%	3.2	Gage
Minor					
201	Presence of flash		0.65%	3.12	Visual
202	Evidence of Poor workmanship		0.65%	3.12	Visual

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QUALITY CONFORMANCE INSPECTION

MIL-M-48186(AR)
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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Overlay, Welded (Prior to welding)	1 OF 1	9298421
			NEXT HIGHER ASSEMBLY 9298412-1

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor 201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Overlay Assembly (Prior to trim of polar hole)	1 OF 1	9298412-1
			NEXT HIGHER ASSEMBLY: 9298405

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Liquid propellant on exterior of overlay		100%	3.2	Visual
102	Quantity of liquid propellant incorrect		0.40%	3.2	Balance Note 1
103	Liquid propellant formulation incorrect				Note 2

Note:

1. The quantity of liquid propellant shall be determined by weight difference of the overlay before and after loading of the propellant.
2. The liquid propellant shall be lotted and a Propellant Description Sheet (AMC Form 1047) shall be prepared for each lot. The propellant shall comply with MIL-P-48333.
3. Alternate: Machine control. Liquid propellant dispensing machines will be qualified at beginning of each production interval, each production hour, and each time the machine is removed from production for maintenance or adjustment. A minimum of two (2) samples will be dispensed and weighted for qualification. Each hour's production or fraction thereof will be accepted or rejected based upon the qualification samples.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Mine (prior to casting or assembly of plug)	1 OF 1	9298400
			NEXT HIGHER ASSEMBLY: 9298400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance from apex to top of kill mechanism and overlay assembly, maximum		0.40%	3.2	Gage
102	Booster of detonator cord assembly protruding beyond keeper in S&A		0.40%	3.2	Visual
103	Keeper not seated on booster		0.40%	3.2	Visual
104	Bedding compound improper		0.40%	3.2	4.5.11
105	Retaining clips missing or incorrectly positioned		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-M-48186A(AR)
AMENDMENT 8

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.26	Mine, Antipersonnel				9298400
					NEXT HIGHER ASSEMBLY: N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shorting bar broken or damaged		100%	3.2	Visual
Major					
101	Evidence of liquid propellant leakage		100%	3.2	Visual
102	Battery ball below minimum		0.40%	3.2	Gage
103	Shorting bar height below minimum		0.40%	3.2	Gage
104	Total height		0.40%	3.2	Gage
105	Arc face radius		0.40%	3.2	Gage
106	Distance from arc face to apex gaging radius		0.40%	3.2	Gage
107	Perpendicularity of arc face to triangular faces		0.40%	3.2	Gage
108	Surface finish on arc face		0.40%	3.2	Gage
109	Any sensor deployed		0.40%	3.2	Finish Blocks
110	Voids in plug bond area		0.40%	3.2	Visual
111	Any sensor restricted (cap not removed or molding compound flash over sensor cutout)		0.40%	3.2	Visual
112	Temperature control during cure		0.40%	3.10	4.5.13

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.26 Cont'd.	Mine, Antipersonnel				9298400
					NEXT HIGHER ASSEMBLY: N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Resin/hardener weight ratio of molding compound		0.40%	3.2	4.5.12
114	Compressive strength of molding compound		0.40%	3.2	4.5.12
115	Molding compound shelf life expired		0.40%	3.2	Visual
116	Arc face color incorrect		0.40%	3.11	Visual
117	Sealing compound not at proper level		0.40%	3.2	Visual
Minor					
201	Test pads exposed		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: Data Item Description DE-R-1721 applies to Critical and 100% Major Characteristics. Submission to the report shall be on a weekly frequency.

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MIL-M-48186A(AR)
AMENDMENT 8

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Sleeve, outer	1 OF 1	9298356
			NEXT HIGHER ASSEMBLY: --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Sleeve, length		0.40%	3.2	Gage
102	Sleeve, inner diameter		0.40%	3.2	Gage
103	Sleeve, outer diameter		0.40%	3.2	Gage
104	Sleeve concentricity		0.40%	3.2	Gage
105	Sleeve, chamfer		0.40%	3.2	Gage
106	Sleeve, hardness	---		3.2	Certification
107	Sleeve, finish		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.28	Explosive Composition A5, prior to loading				9298358
					NEXT HIGHER ASSEMBLY
					9298358
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bulk density			3.7	4.5.2.1
102	Moisture content			3.7	4.5.1
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

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